

SAKSHAM SINGH SACHAN

B.Tech Computer Science and Engineering Student

[Gmail](#) | [GitHub](#) | [LinkedIn](#) | [Twitter](#) | [Leetcode](#) | [Codeforces](#) | [Website](#)

Education

B.Tech in Computer Science and Engineering Present
CSJMU UIET(formerly Kanpur University)

Achievements

- Lichess rating of approximately **2300-2400**
- Chess.com rating of approximately **1930-2000**.

Technical Skills

Programming Languages: JavaScript, TypeScript, Rust, Python, C/C++, Java, Bash, Dart/Flutter, HTML/CSS
Frameworks and Tools: React, Svelte, Astro, Tailwind CSS, Node.js, Express, SQL/NoSQL (MongoDB), Docker, Kubernetes, Git, NGINX, Caddy

Cybersecurity: Offensive Security, Application Security (Web/API), Vulnerability Research, Threat Intelligence, OSINT, Attack Surface Management (ASM), Threat Hunting, Secure Code Review, Authentication and Authorization Testing, Business Logic Testing, API Security, IDOR and Access Control Analysis, Reconnaissance Automation, Detection Engineering, MITRE ATT&CK, Burp Suite, Nmap, ffuf, Nuclei, Metasploit, Wireshark, Ghidra

Operating Systems: Linux (Arch, Debian, Ubuntu, Kali), Windows, Android, Embedded Linux; Proficient with Linux internals, system administration, networking, shell scripting, virtualization (Docker/UTM), and CLI-based development workflows.

DevOps and Infra: GitHub Actions, Fly.io, DigitalOcean, GCP, CI/CD, Railway, Heroku

Database and ORMS MongoDB, SQLite, Firebase, MySQL, Postgres, Drizzle, Prisma

Areas of Interest: Web Application Security, Ethical Hacking, Vulnerability Research, Security Automation

Certifications

- **Arch1001: x86-64 Assembly** – OpenSecurityTraining2 *Issued Jul 2026*
- **Ethical Hacker** – Cisco *Issued Jun 2026*
- **Introduction to Cybersecurity** – Cisco *Issued Jun 2026*

Security Research & Vulnerability Disclosure

Vulnerability Disclosure – IIT Kanpur Website

- Identified and responsibly disclosed a security vulnerability affecting an IIT Kanpur web property through independent security research.
- Conducted security testing and analysis to identify the vulnerability and responsibly communicated the finding to the relevant stakeholders.
- The vulnerability report was acknowledged by an IIT Kanpur professor, demonstrating responsible disclosure and recognition of the security finding.

Physics & Space Science

Research Interests: Physics, Astrophysics, Space Science, Earth Observation, Atmospheric Science, Computational Physics

Selected for ISRO Training Programmes

- **AI/ML for Geodata Analysis** *ISRO — Aug 03–14, 2026*
- **Earth Observations and Numerical Model Applications for Tropical Cyclone Monitoring and Forecasting** *ISRO — Jul 30, 2026*
- **Aerosols: Measurement, Retrieval and Impacts** *ISRO — Jul 27–28, 2026*

Interests: Exploring the intersection of physics, computational methods, artificial intelligence, and space science, with a particular interest in astrophysics, atmospheric phenomena, and Earth observation.

Projects

SSOS – Custom Operating System

Systems Programming — C/C++

- Developing a custom operating system from scratch to gain hands-on experience in low-level systems programming, computer architecture, and kernel development.
- Working on core operating-system concepts including kernel development, memory management, process management, interrupts, system calls, and hardware-level interaction.
- Exploring the complete software stack from system boot and low-level hardware interaction to higher-level operating-system abstractions.

Website Security Scanner

Python — Cybersecurity

- Developed a Python-based security scanning tool to automate website reconnaissance and identify potential security weaknesses.
- Implemented automated security checks and reconnaissance functionality to streamline the initial assessment of web applications.
- Designed the project to strengthen practical understanding of web application security, HTTP communication, vulnerability assessment, and security automation.

Instagram OSINT Tool

Python — OSINT — Open Source

- Developed a Python-based OSINT tool focused on gathering and analysing publicly available information associated with Instagram profiles.
- Automated repetitive information-gathering workflows to improve the efficiency of open-source intelligence investigations.
- Designed the project to explore practical OSINT methodologies, data collection, and Python-based automation, and published it as an open-source GitHub project.

AI-Based Resume Shortlisting System

AI — Python — Web Application

- Developed a prototype AI-powered resume shortlisting system designed to assist recruiters in analysing candidate resumes and identifying relevant applicants.
- Implemented an automated resume-analysis workflow to evaluate candidate profiles against recruitment requirements and support the shortlisting process.
- Built a functional prototype with an interactive user interface and integrated AI functionality, with the potential for further development into a complete recruitment platform.

Rocket Landing Simulator

Python — Simulation — Physics

- Developed a rocket landing simulation inspired by real-world autonomous rocket landing systems, focusing on the computational modelling of a controlled landing trajectory.
- Implemented simulation logic to model rocket motion and analyse the effect of control parameters on the landing process.
- Explored concepts related to physics-based simulation, trajectory control, and autonomous landing systems through an interactive computational environment.

Cybersecurity Experience

Hands-on Cybersecurity Practice

- Practicing cybersecurity through hands-on challenges involving Web Security, API Security, OSINT, Reverse Engineering and Operating Systems.
- Working with platforms including PortSwigger Web Security Academy, pwn.college and picoCTF.
- Developing practical knowledge of reconnaissance, Linux systems, web application vulnerabilities and security testing methodologies.

Bug Bounty and Vulnerability Research

- Learning real-world web application security testing and responsible vulnerability disclosure methodologies.
- Studying reconnaissance, asset discovery, vulnerability identification and security testing workflows.
- Exploring tools including Burp Suite, Nmap, Shodan and Linux-based security utilities.

Open Source

- Maintaining a GitHub portfolio focused on cybersecurity, Python development, OSINT and security automation.
- Building practical cybersecurity projects and security-focused utilities.
- Interested in contributing to open-source cybersecurity projects.

Activities

- Creating beginner-friendly cybersecurity content covering CTF walkthroughs and cybersecurity concepts.
- Developing faceless short-form educational content for students interested in cybersecurity.
- Exploring cybersecurity, OSINT, open-source development and security research.

Relevant Coursework

Data Structures and Algorithms
Database Management Systems
Web Technologies

Computer Networks
Computer Architecture
Software Engineering

Operating Systems
Object-Oriented Programming
Cybersecurity Fundamentals